



COURSE DESCRIPTION

AI has moved beyond research and experimentation into core business operations. Advances in deep learning, reinforcement learning, and generative models are automating routine tasks while augmenting complex work, with measurable productivity gains across sectors. The economic projections are substantial. McKinsey estimates AI could add \$13 trillion to global GDP by 2030;¹ Goldman Sachs puts the figure at \$7 trillion.² The actual number is uncertain, but firms are integrating AI into products, services, and workflows at an accelerating pace. Labor markets are shifting in parallel. Most occupations will see at least some tasks transformed, and employers are paying wage premiums for AI-related skills. Whether automation creates net job growth remains debated, but the mismatch between demand and supply of trained professionals is already apparent. One pattern stands out in the data: many companies have yet to realize measurable benefits from AI investments. The technology alone is insufficient—organizational redesign and workforce readiness are proving equally important.

This course provides a systematic introduction to recent developments in AI, covering foundational concepts in neural networks, business applications, and hands-on experience with deep neural networks. Beyond technical fluency, the course addresses organizational and strategic dimensions that determine the success of AI initiatives.

Upon completion of this course, students will:

1. Understand the fundamentals of AI, e.g., what it can do and cannot do, and the foundations of neural networks;
2. Gain deep knowledge of modern deep learning, including various deep neural network (NN) architectures (MLPs, CNNs, Transformers, LLMs) and other state-of-the-art applications, including agentic frameworks.
3. Be able to apply AI-based methods to solve real-world business problems, understanding considerations for the integration of AI into an existing business process.

REQUIRED TEXTBOOK & READING MATERIALS

Textbooks

Required Textbook:

- Chollet, François. (2025). *Deep Learning with Python (3rd Edition)*. Manning Publications Co. **ISBN-13: 1633436589**. <https://www.manning.com/books/deep-learning-with-python-third-edition>

Optional Readings:

- *Artificial Intelligence: 101 Things You Must Know Today About Our Future*. Lasse Rouhiainen, CreateSpace Independent Publishing Platform, 2018.
- *Deep Learning*. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, MIT Press, 2016

¹ [McKinsey Global Institute – Notes from the AI Frontier: Modeling the Impact of AI on the World Economy](#).

² [Goldman Sachs – Generative AI Could Raise Global GDP by 7%](#).

Business Cases:

- Bojinov, Iavor, Karim Lakhani, and David Lane (2025). *JPMorganChase: Leadership in the Age of Gen AI*. Harvard Business School Case #325066-PDF-ENG.
- Wu, Andy, Matt Higgins, Miaomiao Zhang, Hang Jiang (2024). *AI Wars* (Revised Feb 12, 2024). Harvard Business School Case #723434-PDF-ENG.

Other Reading Materials:

These reading materials (extracted PDFs from websites) will be available on NYU Brightspace. You can either read PDF files or access the links provided below:

Schedule of Required Readings:

Pre-Module Readings (must read prior to the first Zoom session):

Note: your pre-module quiz will cover Chapters 1, 2, and 3 of the textbook.

- You are required to read Chapters 1, 2, and 3 of the required textbook in detail and try to implement the code in Chapter 2.
- You are also required to have read the AI Wars and JPMorgan Chase cases prior to the in-class module.

Optional Pre-module Reading:

- Neural Networks and Deep Learning (Chapter 1) by Michael Nielsen: <http://neuralnetworksanddeeplearning.com/chap1.html>
- A Complete History of AI: <https://learn.g2.com/history-of-artificial-intelligence>

Post-Module Readings:

Note: Chapters 5 and 6 of the required book will strengthen your understanding of some topics that we cover in class. You can also reference the reading materials below after the class has completed, if you have time and if you want to dive deeper into these topics.

- Convolutional Neural Networks (CNN) for Visual Recognition: <https://cs231n.github.io/convolutional-networks/>
- AI Transformation Playbook (see the uploaded PDF)
- Understanding LSTM networks: <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>
- Tutorial on Generative Adversarial Networks (GANs): <https://medium.com/ai-society/gans-from-scratch-1-a-deep-introduction-with-code-in-pytorch-and-tensorflow-cb03cdcd8a0f>

SOFTWARE

We will use the Keras library in Python. To use the code examples and follow along, you will need to create a Google Colab account (<https://colab.research.google.com/>).

Note that using Colab will be much easier than using your own computer because the virtual machines Colab provides access to provide GPUs, which will be important for fitting models on larger datasets. Note as well that students now have free access to a Colab Pro subscription, once you validate a .edu email address via the Google Colab subscriptions page. And, Google Colab provides native integrated access to Google Gemini 3, which can aid in understanding the coding examples and in producing new code implementations.

For the Post module exercise, you are expected to modify the Colab Notebooks provided in class to accomplish your task.

Optional Software That I Recommend You Explore:

- Claude Code + Chrome Browser Extension: <https://www.youtube.com/watch?v=AJpK3YTTKZ4>

GRADING

Grading is based on the pre-module quiz on NYU Brightspace (25%) and the post-module assignment (75%). Below is the grading guideline defined by the Stern School:

- 30-35% of the grades will be A.
- 50-70% of the grades will be B.
- 5-15% of the grades will be C or lower.

COURSE SCHEDULE

Pre-Module

1. Complete all the pre-module reading materials (see Reading Materials Section above). Complete Chapters 1 through 3 of the book **before the Pre-Module Quiz deadline**.
2. Complete the pre-module quiz based on the required pre-module readings (Chapters 1, 2, and 3 of the textbook). **The quiz must be completed by Wednesday, February 18, at 5:00 pm ET.**
3. Create a Google account and make sure you can access Google Colab. Register for a Google Colab Pro subscription (free educational access). You can do this via the settings menu in Google Colab (click the gear icon, then choose Colab Pro, and click the bottom button re: educational access).
4. Read the AI Wars and JPMorganChase HBS cases before the in-class module.

Pre-Module Zoom Session 1 (Intro and Overview) Wednesday, February 18, 6:30 – 8:00 pm ET

Readings: Chapters 1, 2, and 3

1. Course overview and introductions
2. What is AI? History and evolution of AI

3. The math behind a Multilayer Perceptron (MLP): vectors, matrices, derivatives, activation functions
4. Loss functions and the backpropagation algorithm

Pre-Module Zoom Session 2 (Math of NN ctd. + Examples of Feedforward NNs)
Wednesday, March 11, 6:30 – 8:00 pm ET

Readings: Chapter 4

1. Building your first neural network: Gradient tape (auto differentiation), optimizers, Keras APIs
2. Training a neural network: Forward pass, Loss, Backpropagation
3. Model tuning: Evaluating model performance, validation techniques
4. Examples of prediction using neural networks

In-Class Module

DAY 1 – Monday, March 23, 2026

Session 1 (8:30 – 10:30 am): Architectures for Image Data (Convolutional NNs)

Readings: Chapter 8

1. **Anatomy of an image or video**
2. **Introduction to Convolutional Neural Networks (CNNs)**
3. **Components and architecture: convolution, stride, pooling layers**
4. **Implementing a CNN for image classification**

Break (10:30 – 11:00 am)

Session 2 (11:00 am – 12:30 pm): Advanced CNNs & Transfer Learning

Readings: Chapters 9 and 10

1. **Advanced CNN architectures: ResNet, Inception**
2. **Techniques for improving CNN performance: batch normalization, separable convolutions**
3. **Transfer learning: utilizing pre-trained image models**
4. **Understanding what ConvNets “see” (inspecting feature maps)**

Lunch (12:30 – 1:30 pm)

Session 3 (1:30 – 3:30 pm): Architectures for Text & Audio (Recurrent NNs and the Transformer)

Readings: Chapters 14 and 15

1. **Working with sequence data (text, audio)**
2. **Creating and working with embeddings (Word2Vec, BERT)**
3. **Introduction to attention mechanisms: dot-product attention**
4. **The Transformer architecture**

DAY 2 – Tuesday, March 24, 2026

Session 4 (8:00 – 10:00 am): Architecture for Generative AI (Large Language Models)

Readings: Chapter 16

1. **Architecture of autoregressive large language models (LLMs)**
2. **Non-determinism via “temperature.”**
3. **Reinforcement Learning with Human Feedback (RLHF)**
4. **Fine-tuning Pre-trained LLMs: Prefix Fine-tuning & Low Rank Adaptation (LoRA)**

Break (10:00 – 10:15 am)

Session 5 (10:15 am – 11:15 pm): Scaffolding for Generative Text Models I

1. **Prompting Strategies & Reasoning Models**
2. **Retrieval Augmented Generation (RAG)**
3. **Quantization & Distillation of LLMs**

Brunch (11:15 – 12:15 pm)

COMPANY VISIT (12:15 – 2:45 pm)

Session 6 (2:45 – 4:15 pm): Scaffolding II (Agentic Frameworks) & Deployment Considerations for Gen AI

Readings: HBS Case – Generative AI at JPMC

1. **Agentic Frameworks**
2. **Understanding ideal and problematic use cases for Generative AI**
3. **Information security, governance, and process design with LLMs and Agents**

Break (4:15 – 4:30 pm)

Session 7 (4:30 – 5:30 pm): Generative AI Value Chain & Product Strategy

Readings: HBS Case – AI Wars

1. **Generative AI value chain**
2. **Business strategy with Generative AI products**

Post-Module Exercise

First, read the post-module reading materials (see Reading Materials Section). Then, complete the following two post-module assignment items:

1. The first part of the assignment is to be done ***independently***. Discussion is allowed, but each student must submit their own individual write-up. Please submit a single edited .ipynb file (with your name and ID in the file name). (40% of final grade).
2. The second part is an open-ended (group) project; you can do it independently or collaborate with up to two other students (a maximum of three students per group). Please submit a separate single WORD/PDF file with all group members' names and IDs. (35% of final grade).

INSTRUCTOR'S BIOGRAPHICAL PROFILE

Dr. Gordon Burtch is the Allen and Kelli Questrom Professor of Information Systems at Boston University's Questrom School of Business and a Research Scientist with Meta's Central Applied Science group. His research focuses on the economic evaluation of information systems and the societal consequences of technology, employing a variety of empirical methods, including econometrics, field experimentation, and machine learning, to identify and quantify the drivers of individual participation in online social contexts. His work also focuses on methodological development at the intersection of causal methods and machine learning/artificial intelligence. Gordon's work has been published in various leading peer-reviewed academic journals, spanning academic disciplines, including the Proceedings of the National Academy of Sciences, Science Advances, Management Science, Marketing Science, the American Journal of Public Health, Information Systems Research, MIS Quarterly, Manufacturing & Service Operations Management, Organization Science, Production and Operations Management, the Journal of Law, Economics & Organization, and the Journal of Consumer Psychology.

Gordon is a recipient of Early Career Awards from the Association for Information Systems (2017) and the Information Systems Society (ISS) of the Institute for Operations Research and Management Science (INFORMS) (2017). Gordon's research has been supported by more than \$2 million in grants from various corporate, non-profit, and government organizations, including the NSF, Ewing Marion Kauffman Foundation, the 3M Foundation, Adobe, Meta, and the European Commission. His work and opinions have been cited by numerous outlets in the popular press, including the Wall Street Journal, NPR, Time Magazine, Forbes, Vice, Wired, the LA Times, Pacific Standard, and PC Magazine. Gordon presently serves as an Associate Editor for two journals: Management Science and ISR.

Prior to his time at BU and Meta, Gordon worked as a consulting researcher at Microsoft. Prior to entering academia, Gordon was employed as an information systems auditor, a hardware design engineer, and a technology consultant with Accenture Canada in Toronto. Gordon teaches graduate and undergraduate courses on data analytics, machine learning, and digital business strategy. He holds a Bachelor of Engineering and a Master of Business Administration from McMaster University, and a PhD in Business Administration from Temple University's Fox School of Business.

ACADEMIC INTEGRITY

Our undergraduate Academics Pillar states that ***we take pride in our well-rounded education and approach our academics with honesty and integrity***. Indeed, integrity is critical to all that we do here at NYU Stern. As members of our community, all students agree to abide by the NYU Academic Integrity Policies as well as the NYU Stern Student Code of Conduct, which includes a commitment to:

- Exercise integrity in all aspects of one's academic work, including, but not limited to, the preparation and completion of exams, papers, and all other course requirements, by not engaging in any method or means that provides an unfair advantage.
- Clearly acknowledge the work and efforts of others when submitting written work as one's own. Ideas, data, direct quotations (which should be designated with quotation marks), paraphrasing, creative expression, or any other incorporation of the work of others should be fully referenced.
- Refrain from behaving in ways that knowingly support, assist, or in any way attempt to enable another person to engage in any violation of the Code of Conduct. Our support also includes reporting any observed violations of this Code of Conduct or other School and University policies that are deemed to adversely affect the NYU Stern community.

The Stern Code of Conduct and Judiciary Process applies to all students enrolled in Stern courses. For graduate students, information can be found here: <https://www.stern.nyu.edu/uc/codeofconduct>. For undergraduates, information can be found here: <https://www.stern.nyu.edu/portal-partners/current-students/undergraduate/community/community-expectations>.

To help ensure the integrity of our learning community, prose assignments you submit to NYU Brightspace will be submitted to Turnitin. Turnitin will compare your submission to a database of prior submissions to Turnitin, current and archived Web pages, periodicals, journals, and publications. Additionally, your document will become part of the Turnitin database.

GENERAL CONDUCT & BEHAVIOR

Students are also expected to maintain and abide by the highest standards of professional conduct and behavior. Please familiarize yourself with Stern's Policy in Regard to In-Class Behavior & Expectations (<https://www.stern.nyu.edu/portal-partners/registrar/policies-procedures/general-policies/code-conduct>) and the NYU Student Conduct Policy (<https://www.nyu.edu/about/policies-guidelines-compliance/policies-and-guidelines/university-student-conduct-policy.html>).

GRADING GUIDELINES

Grading Information for Stern Core Courses

At NYU Stern, we strive to create courses that challenge students intellectually and that meet the Stern standards of academic excellence. To ensure fairness and clarity of grading, the Stern faculty has adopted a grading guideline for core courses with enrollments of more than 25 students, in which approximately 35% of students will receive an "A" or "A-" grade. There is no enrollment minimum for graduate courses, but this rule applies to the Undergraduate core with enrollments of more than 25 students. In Undergraduate core classes with fewer than 25 students, the instructor is at liberty to assign whatever grades they believe the students deserve while maintaining rigorous academic standards.

Grading Information for Stern Elective Courses

At NYU Stern, we strive to create courses that challenge students intellectually and that meet the Stern standards of academic excellence. To ensure fairness and clarity of grading, the Stern faculty have agreed that for elective courses, the individual instructor or department is responsible for determining reasonable grading guidelines.

STUDENT ACCESSIBILITY

If you will require academic accommodation of any kind during this course, you must notify me at the beginning of the course and provide a letter from the Moses Center for Student Accessibility (212-998-4980, mosescsa@nyu.edu) verifying your registration and outlining the accommodations they recommend. If you will need to take an exam at the Moses Center for Student Accessibility, you must submit a completed Exam Accommodations Form to them at least one week prior to the scheduled exam time to be guaranteed accommodation. For more information, visit the CSA website: <https://www.nyu.edu/students/communities-and-groups/student-accessibility.html>

STUDENT WELLNESS

Our aim is for students to be as academically successful as possible and to help them overcome any impediments to that success. Bookmark the NYU Stern Well-being Resource Hub (<https://www.stern.nyu.edu/wellbeing>) for existing services at NYU and Stern covering a wide range of topics, including financial, relationship, and mental well-being. Any student who may be struggling and believes this may affect their performance in this course is urged to contact the Moses Center for Student Accessibility at 212-998-4980 to discuss academic accommodations. If mental health assistance is needed, call NYU's 24/7 Wellness Exchange hotline at 212-443-9999. Furthermore, please feel free to approach me if you are comfortable doing so. This will enable me to provide relevant resources or referrals. There are also drop-in hours and appointments. Find out more at <http://www.nyu.edu/students/health-and-wellness/counseling-services.html>

NAME PRONUNCIATION AND PRONOUNS

NYU Stern students now can include their pronouns and name pronunciation in Albert. I encourage you to share your name pronunciation and preferred pronouns this way. Please utilize this link for additional information:

<https://www.nyu.edu/students/student-information-and-resources/registration-records-and-graduation/forms-policies-procedures/pronouns-and-name-pronunciation.html>

RELIGIOUS OBSERVANCES AND OTHER ABSENCES

NYU's Calendar Policy on Religious Holidays states that members of any religious group may, without penalty, absent themselves from classes when required in compliance with their religious obligations. You must notify me in advance of religious holidays or observances that may coincide with exams, assignments, or class times so we can schedule mutually acceptable alternatives. Students may also contact religiousaccommodations@nyu.edu for assistance.

Except for religious observances or other absences required by nondiscrimination law, this class requires attendance and participation and cannot accommodate conflicts. Please review all class dates at the start of the semester and review all course requirements to identify any foreseeable conflicts with exams, course assignments, projects, or other items required for participation and attendance. If you are aware of a potential conflict, it is strongly recommended that you do not take this class.

LAPTOPS, CELL PHONES & OTHER ELECTRONIC DEVICES

The use of electronic devices (e.g., tablets or laptops), for the purpose of notetaking and in-class exercises, is permitted. However, students should make every effort to avoid distracting their classmates or disrupting the class, including arriving early and choosing a seat that is less distracting for peers.

INCLUSION STATEMENT

This course strives to support and cultivate diversity of thought, perspectives, and experiences. The intent is to present materials and activities that will challenge your current perspectives with a goal of understanding how others might see situations differently. By participating in this course, everyone is expected to commit to making it an inclusive learning environment for all.

GENERATIVE AI POLICY

Students in this course are expected and encouraged to use generative AI tools, crediting them when used, even if only to generate ideas. When you submit your assignment, you must attribute what text was generated by the AI tool (e.g., include an in-text parenthetical citation if AI text has been paraphrased). When using AI tools on assignments, you can add an appendix showing (a) the entire exchange, highlighting the most relevant sections; (b) a description of precisely which AI tools were used, (c) an explanation of how the AI tools were used (e.g., to generate ideas, turns of phrase, elements of text, long stretches of text, lines of argument, pieces of evidence, maps of the conceptual territory, illustrations of key concepts, etc.); (d) an account of why AI tools were used (e.g., to save time, to surmount writer's block, to stimulate thinking, to handle mounting stress, to clarify prose, to translate text, to experiment for fun, etc.). Overall, AI tools should be used wisely and reflectively to deepen understanding of the subject matter.